# Vectors I: Homework 2

Compiled by Maximus Pung Jun

Scalar product: properties, angles, perpendicularity, projections, ratios. Q1 True/False from the Basics Worksheet (statements (e) and (p) omitted). Q2–Q7 are past-paper selections on the scalar product. Total: 42 marks (Q1 unmarked).

## Questions

### 1

For the following statements, state whether it is **true** or **false**.

| | **Statement** | **T/F** |
|---|---|---|
| **(a)** | $\mathbf{a}\cdot\mathbf{b}=\mathbf{b}\cdot\mathbf{a}$. | |
| **(b)** | $\mathbf{a}\cdot\mathbf{a}=\lvert\mathbf{a}\rvert^2$. | |
| **(c)** | $\mathbf{i}\cdot\mathbf{j}=0$. | |
| **(d)** | $\mathbf{a}\cdot\mathbf{b}$ is a vector. | |
| **(f)** | If $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$, then $\mathbf{a}\cdot\mathbf{b}=0$. | |
| **(g)** | $(2\mathbf{a})\cdot(3\mathbf{b})=6\,\mathbf{a}\cdot\mathbf{b}$. | |
| **(h)** | $(\mathbf{a}\cdot\mathbf{b})\mathbf{c}$ is a vector. | |
| **(i)** | $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$, where $\theta$ is the non-reflex angle between $\mathbf{a}$ and $\mathbf{b}$. | |
| **(j)** | $\begin{pmatrix}1\\2\\3\end{pmatrix}\cdot\begin{pmatrix}2\\3\\5\end{pmatrix}=\begin{pmatrix}2\\6\\15\end{pmatrix}$. | |
| **(k)** | If $\mathbf{a}\cdot\mathbf{b}=0$, then $\mathbf{a}=\mathbf{0}$ or $\mathbf{b}=\mathbf{0}$. | |
| **(l)** | $\lvert\mathbf{a}\cdot\mathbf{b}\rvert=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. | |
| **(m)** | $\mathbf{a}\cdot\mathbf{b}\le\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$. | |
| **(n)** | For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}>0$ means that the angle $\theta$ between $\mathbf{a}$ and $\mathbf{b}$ is acute or zero ($0^\circ\le\theta<90^\circ$). | |
| **(o)** | For non-zero $\mathbf{a}$ and $\mathbf{b}$, $\mathbf{a}\cdot\mathbf{b}<0$ means that the angle $\theta$ between $\mathbf{a}$ and $\mathbf{b}$ is obtuse or $180^\circ$ ($90^\circ<\theta\le180^\circ$). | |
| **(q)** | $\mathbf{a}\cdot(\mathbf{b}\cdot\mathbf{c})$ is a scalar. | |
| **(r)** | $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}+\mathbf{b})=\lvert\mathbf{a}\rvert^2+\lvert\mathbf{b}\rvert^2$. | |
| **(s)** | For non-zero $\mathbf{a}$ and $\mathbf{b}$, if $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert$, then $\mathbf{a}$ and $\mathbf{b}$ are parallel. | |
| **(t)** | $\mathbf{a}\cdot\mathbf{b}=\lvert\mathbf{a}\rvert\lvert\mathbf{b}\rvert\cos\theta$ also holds if $\theta$ is the reflex angle between $\mathbf{a}$ and $\mathbf{b}$. | |
| **(u)** | Let $\theta$ be the angle $ABC$. Then $\overrightarrow{BA}\cdot\overrightarrow{BC}=\lvert\overrightarrow{BA}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$. | |
| **(v)** | The length of projection of $\mathbf{a}$ onto $\mathbf{b}$ is equal to the length of projection of $\mathbf{b}$ onto $\mathbf{a}$. | |
| **(w)** | If $\lvert\mathbf{a}\rvert=\lvert\mathbf{b}\rvert$, then $(\mathbf{a}+\mathbf{b})\cdot(\mathbf{a}-\mathbf{b})=0$. | |
| **(x)** | $\mathbf{a}\cdot\mathbf{b}=\mathbf{a}\cdot\mathbf{c}$ means that $\mathbf{b}=\mathbf{c}$. | |
| **(y)** | Let $\theta$ be the angle $ABC$. Then $\overrightarrow{AB}\cdot\overrightarrow{BC}=\lvert\overrightarrow{AB}\rvert\lvert\overrightarrow{BC}\rvert\cos\theta$. | |
| **(z)** | Let $\theta$ be the angle $CBD$. Then $\overrightarrow{BD}\cdot\overrightarrow{CB}=\lvert\overrightarrow{BD}\rvert\lvert\overrightarrow{CB}\rvert\cos\theta$. | |

**2** [EJC 2024 Promo Q5]

$\mathbf{a}$ and $\mathbf{b}$ are vectors such that $|\mathbf{a}|=\sqrt{2}$ and $|\mathbf{b}|=\sqrt{6}$. The angle between $\mathbf{a}$ and $\mathbf{b}$ is $\frac{5\pi}{6}$.

**(a)** Find the exact length of projection of $\mathbf{a}$ onto $\mathbf{b}$. **[2]**

**(b)** By considering $(3\mathbf{a}+2\mathbf{b})\cdot(3\mathbf{a}+2\mathbf{b})$, find the exact value of $|3\mathbf{a}+2\mathbf{b}|$. **[4]**

**3** [SAJC 2024 Promo Q9(a)]

Relative to the origin $O$, the points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively, where $\mathbf{a}$ and $\mathbf{b}$ are non-parallel. It is given that $|\mathbf{a}|=9$, $|\mathbf{b}|=1$ and $|2\mathbf{a}+\mathbf{b}|=2\sqrt{74}$.

**(i)** By considering $(2\mathbf{a}+\mathbf{b})\cdot(2\mathbf{a}+\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=-\frac{29}{4}$. **[3]**

**(ii)** Give a geometrical meaning of $|\mathbf{a}\cdot\mathbf{b}|$. **[1]**

**(iii)** The points $P$, $Q$ and $R$ have position vectors $7\mathbf{a}-5\mathbf{b}$, $6\mathbf{a}+5\mathbf{b}$ and $9\mathbf{a}+\lambda\mathbf{b}$ respectively. Given that these three points are collinear, find the value of $\lambda$. **[3]**

**4** [CJC 2022 Promo Q9(i),(ii)]

Relative to the origin $O$, the position vectors of points $A$, $B$ and $C$ are $\mathbf{a}$, $\mathbf{b}$ and $\mathbf{c}$ respectively, where $\mathbf{b}$ is a unit vector. It is given that $\mathbf{b}$ and $\mathbf{b}-\mathbf{a}$ are perpendicular and $C$ lies on $AB$ such that $AC:CB=3:1$.

**(i)** Show that $|\mathbf{a}|=\sec\theta$, where $\theta$ is the angle between $\mathbf{a}$ and $\mathbf{b}$. **[3]**

By expressing $\mathbf{c}$ in terms of $\mathbf{a}$ and $\mathbf{b}$,

**(ii)** find the value of $|\mathbf{c}\cdot\mathbf{b}|$ and state the geometrical interpretation of $|\mathbf{c}\cdot\mathbf{b}|$. **[4]**

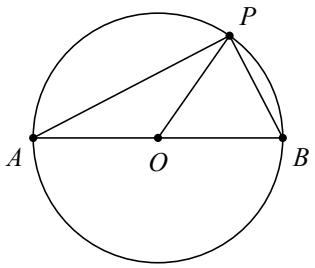
**5** [SAJC 2025 Promo Q8(i)]

Relative to the origin $O$, two points $A$ and $B$ have position vectors $\mathbf{a}$ and $\mathbf{b}$ respectively. It is given that $|\mathbf{a}|=2$, $|\mathbf{b}|=1$ and $|3\mathbf{a}-2\mathbf{b}|=\sqrt{37}$.

**(i)** By considering the scalar product $(3\mathbf{a}-2\mathbf{b})\cdot(3\mathbf{a}-2\mathbf{b})$, show that $\mathbf{a}\cdot\mathbf{b}=\frac{1}{4}$, and give the geometrical meaning of $|\mathbf{a}\cdot\hat{\mathbf{b}}|$. **[4]**

**6** [EJC 2025 Prelim P1 Q6]

There are three distinct points $A$, $B$ and $P$. The point $P$ lies on the circle with centre $O$ and diameter $AB$. Relative to the origin $O$, the position vectors of points $A$ and $P$ are **a** and **p** respectively.




**(a)** By expressing $\overrightarrow{AP}$ and $\overrightarrow{BP}$ in terms of **a** and **p**, show that $\overrightarrow{AP}\cdot\overrightarrow{BP}=0$. **[3]**

The point $C$ with position vector **c** lies on $AB$ produced such that $AC:AB$ is $\lambda:1$.

**(b)** Find **c** in terms of $\lambda$ and **a**. **[2]**

**(c)** It is given that $PC$ is a tangent to the circle and that angle $AOP$ is $120°$. Using a suitable scalar product, find the value of $\lambda$. **[4]**

**7** [NJC 2025 Prelim P2 Q3]

With respect to an origin $O$, points $P$ and $Q$ have variable position vectors **p** and **q** respectively, given by

$$\mathbf{p}=(\cos t)\,\mathbf{i}+(\sin t)\,\mathbf{j}-\mathbf{k}, \qquad \mathbf{q}=(\cos 2t)\,\mathbf{i}-(\sin 2t)\,\mathbf{j}+\alpha\mathbf{k},$$

where $\alpha$ is a real parameter and $0<t<\pi$.

**(i)** Find the exact values of $t$ and $\alpha$ if $O$, $P$ and $Q$ are collinear. **[5]**

It is given that $\alpha=0.5$.

**(ii)** Show that $\mathbf{p}\cdot\mathbf{q}=\cos 3t-0.5$. Find the greatest length of projection of **q** onto **p**. **[4]**

# Answers

**1** **(a)** T **(b)** T **(c)** T **(d)** F **(f)** T **(g)** T **(h)** T **(i)** T **(j)** F **(k)** F **(l)** F **(m)** T **(n)** T **(o)** T **(q)** F **(r)** F **(s)** T **(t)** T **(u)** T **(v)** F **(w)** T **(x)** F **(y)** F **(z)** F

**2** **(a)** $\frac{\sqrt{6}}{2}$

**(b)** $\sqrt{6}$

**3** **(i)** Shown.

**(ii)** Since $|\mathbf{b}|=1$, $|\mathbf{a}\cdot\mathbf{b}|$ is the length of the projection of **a** onto **b**.

**(iii)** $\lambda=-25$

**4** **(i)** Shown.

**(ii)** $|\mathbf{c}\cdot\mathbf{b}|=1$; $|\mathbf{c}\cdot\mathbf{b}|$ is the length of the projection of **c** onto **b**.

**5** **(i)** Shown. $|\mathbf{a}\cdot\hat{\mathbf{b}}|$ is the length of the projection of **a** onto **b**.

**6** **(a)** Shown.

**(b)** $\mathbf{c}=(1-2\lambda)\mathbf{a}$

**(c)** $\lambda=\frac{3}{2}$

**7** **(i)** $t=\frac{\pi}{3}$, $\alpha=1$ or $t=\frac{2\pi}{3}$, $\alpha=-1$

**(ii)** Shown. Greatest length $=\frac{3}{2\sqrt{2}}$ $(=1.06$ (3 s.f.))

**END OF WORKSHEET**